

PSYC 51.07: Models of Language and Communication

Meeting times	MWF 10:10–11:15
X-hour	Th 12:15–1:05
Classroom	Moore 302
Instructor	Dr. Jeremy R. Manning
Email	jeremy@dartmouth.edu
Office location	349 Moore Hall
Office hours	[By appointment](https://context-lab.youcanbook.me)

Course Information

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Meeting times	MWF 10:10–11:15
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Course Description

Recent advancements in Artificial Intelligence (AI) have led to the development of powerful large language models (LLMs) capable of generating human-like text. These models have revolutionized the way we think about language, cognition, and intelligence. This course introduces students to the various approaches used in building, studying, and using conversational agents. We will use readings, discussions, and hands-on demonstrations to explore historical approaches, current approaches, and hints about the “next generation” of these models.

We will investigate what these models can and cannot do from various cognitive, philosophical, computational, and ethical perspectives. A special focus will be on which aspects of human cognition, experiences, minds, and language these models can vs. cannot capture.

Course Goals

This course is intended to train students to:

- Explore a variety of approaches to constructing conversational agents, including string manipulation models, rules-based (decision tree) models, and transformer-based models. Students will implement and train these models, largely from scratch, to gain a deep understanding of how they work.

- Understand different classic and modern approaches to text embeddings (LSA, LDA, word2vec, USE, BERT, etc.) and dimensionality reduction (PCA, UMAP). Students will use these approaches to carry out analyses of text corpora.
- Gain a general understanding of computational linguistic text analyses including part of speech tagging, sentiment analysis, etc. Again, students will implement and apply these tools to text data.
- Explore the psychological and cognitive applications of LLMs, such as modeling thought processes, language comprehension, and social cognition
- Critically assess the abilities and limitations of conversational agents in replicating human-like reasoning and understanding
- Discuss the philosophical and ethical implications of AI, including questions around consciousness, agency, and the future of AI
- Develop and implement experiments using LLMs to explore research questions in psychology and cognitive science

Pre-Requisites

Students must have prior experience with Python programming in order to enroll in this course. Prior coursework on statistics or probability (e.g., PSYC 10, AP Stats, or similar) is also highly recommended. An online statistics course is fine as a stand-in, but I will expect you to know about hypothesis testing, probability distributions, and have some intuitions about how to design and interpret statistical analysis before you start this course. Additional prior coursework and/or experience with linguistics, linear algebra, statistics, machine learning, artificial intelligence, data science, philosophy of mind, and/or creative writing will all be useful, but are not required.

Course Materials

We will use a variety of freely available online materials and research papers, which will be provided throughout the course. You will also need an internet-enabled computer or tablet capable of displaying and outputting graphics and running a standard web browser (e.g., Chrome or Firefox).

Format and Overview

This course follows an **experiential learning** model. Students will engage with lecture materials through hands-on programming exercises, experiments with models and tools, and group discussions. Problem sets will deepen your understanding of the course material, and small projects will allow you to apply concepts to real-world research problems in the relevant domains.

Classes will include:

- **Lectures:** Foundational topics, discussions on implementing and leveraging conversational agents, text embeddings, and text analysis.
- **Hands-on Labs:** Experimenting with conversational agents, designing simple conversational agents, and conducting research using these models.
- **Problem Sets and Projects:** Bi-weekly problem sets to apply what you've learned. These will typically take the form of small-scale "research" projects.

Platforms and Tools

- **Google Colaboratory:** For developing and running Python code.
- **Hugging Face Models:** We will use pre-trained models available through Hugging Face's model hub.
- **GitHub:** Used for managing and sharing code, data, and project work.
- **Discord:** To facilitate discussions outside of class, share ideas, and collaborate on projects.

Grading

Grades will be based on the following components:

- **Problem Sets (75%):** A total of 5 problem sets designed to reinforce key concepts (each is worth 15% of the final course grade).
- **Final Project (25%):** You will carry out a larger scale (relative to the problem sets) "research" project on a topic of your choosing. This will include:
 - Python code, organized as a Colaboratory notebook
 - A "presentation" to the class (also submitted as a YouTube video), along with an in-class discussion of your project
 - A brief (2–5 page) writeup of the main approach and key findings or takeaways

Students may work together on all of the assignments, unless otherwise noted in class or in the assignment instructions. However, **each student must submit their own problem set and indicate who they worked with.** Final projects will (typically) be completed in groups of 2–3 students, with the entire group turning in the same project (and receiving the same grade for it).

Grading Scale: A (93–100), A– (90–92), B+ (87–89), B (83–86), B– (80–82), C+ (77–79), C (73–76), C– (70–72), D (60–69), E (0–59). All grades will be rounded to the nearest integer (e.g., a 92.5 average will result in a final grade of "A", whereas a 92.4999 average will result in a final grade of "A–"). Out of fairness to all students in the course, there will be no "negotiations" about grading— e.g., your grade will be determined solely by the numerical average of your assignment scores.

Late Policy

Problem sets will receive a 10% deduction for each week late, rounded **up** to the nearest whole week (e.g., from a grading standpoint submitting an assignment 1 minute late is the same as submitting it 1 day late, is the same as submitting it 6 days late).

Your final project **must be submitted on time** in order to receive credit for it.

I strongly encourage you to submit your assignments before the last possible moment to avoid grading penalties, unexpected circumstances (e.g., illness, emergencies, etc.).

Academic Honor Principle

Students are expected to adhere to Dartmouth's Academic Honor Principle. You are encouraged to collaborate and discuss ideas with classmates, but all submitted work must be your own (aside from the final projects, which will be

completed in small groups). If you're unsure about what constitutes a violation, please ask for clarification.

Use of Generative AI

Given that this is a course about the technologies used to make modern generative AIs like ChatGPT, Claude, Llama, Gemini, etc., you will be using these models extensively in class and in your assignments. There are just two "rules" about using GenAI:

- Most importantly, **you** are responsible for the content of your assignments, whether written by you "from scratch" or with the help of a GenAI agent.
- Second, you are bound by the Academic Honor Principle to acknowledge any use of GenAI in your work. This can be done by either using a brief comment in your code, by explicitly citing the tools you use, or by adding a note to the relevant section(s) of your assignment. Each situation is unique, but you need to make it clear exactly what work is your own vs. produced by GenAI. You must also include a chat history (including any prompts you used) as an addendum to your assignment(s).

Scheduling Conflicts

Attendance is expected for all classes unless previously arranged. A critical part of the course is the in-class discussions and demos, and those will only work if you are physically present in class. If you anticipate any conflicts due to religious observances or other commitments, please inform the instructor by Week 2 to make appropriate arrangements.

Student Needs

We strive to create an inclusive learning environment where all students feel supported and engaged. If you require any accommodations, please contact the Student Accessibility Services office, or discuss your needs with the instructor privately.

Course Schedule

Note: Classes meet MWF 10:10–11:15. X-hours (Th 12:15–1:05) in the first three weeks will be used to make up for instructor absence February 23–27 (no classes that week).

Week	Dates	Topics	Assignments
1	Jan 5–9	Introduction, Pattern Matching, ELIZA	Assignment 1 released
2	Jan 12–16	Data Cleaning, Tokenization, POS Tagging	Assignment 2 released, Assignment 1 due (Jan 19, 11:59 PM EST)
3	Jan 19–23	Classic Embeddings (LSA, LDA), Word Embeddings	Assignment 3 released, Assignment 2 due (Jan 26, 11:59 PM EST)

Week	Dates	Topics	Assignments
4	Jan 26–30	Contextual Embeddings, Dimensionality Reduction	Assignment 4 released, Assignment 3 due (Feb 2, 11:59 PM EST)
5	Feb 2–6	Attention Mechanisms, Transformer Architecture	Assignment 4 due (Feb 9, 11:59 PM EST)
6	Feb 9–13	BERT Deep Dive, Encoder Models	Assignment 5 released
7	Feb 16–20	GPT Architecture, Scaling Up	Assignment 5 due (Feb 16, 11:59 PM EST)
8	Feb 23–27	No classes (instructor away)	Work on Final Project
9	Mar 2–6	RAG, Mixture of Experts, Ethics	
10	Mar 9	Final Project Presentations	Final Project due (Mar 9, 11:59 PM EST)
Finals	Mar 13		

Detailed Schedule

Week 1: Introduction & String Manipulation (January 5–9)

Day	Lecture	Topics	Slides	Materials
Mon Jan 5	Lecture 1	Course Introduction, Is ChatGPT Conscious?	HTML, PDF	Fedorenko et al. (2024), Schrimpf et al. (2021)
Wed Jan 7	Lecture 2	Pattern Matching & ELIZA	HTML, PDF	Weizenbaum (1966)
Thu Jan 8	Lecture 3	ELIZA Deep Dive	HTML, PDF	Start Assignment 1
Fri Jan 9	Lecture 4	ELIZA Implementation & The ELIZA Effect	HTML, PDF	Natale (2021), Assignment 1 Released

Week 2: Computational Linguistics (January 12–16)

Day	Lecture	Topics	Slides	Materials
Mon Jan 12	Lecture 5	Data Cleaning & Preprocessing	HTML, PDF	HuggingFace NLP Course Chapter 2
Wed Jan 14	Lecture 6	Tokenization	HTML, PDF	Sennrich et al. (2016), Kudo & Richardson (2018)

Day	Lecture	Topics	Slides	Materials
Thu Jan 15	Lecture 7	Text Classification Workshop	HTML, PDF	
Fri Jan 16	Lecture 8	POS Tagging & Sentiment Analysis	HTML, PDF	Assignment 2 Released, Assignment 1 Due

Week 3: Text Embeddings I (January 19–23)

Day	Lecture	Topics	Slides	Materials
Mon Jan 19	–	Martin Luther King Jr. Day (No Class)	–	
Wed Jan 21	Lecture 9	Classic Embeddings (LSA, LDA)	HTML, PDF	Landauer & Dumais (1997), Blei et al. (2003)
Thu Jan 22	Lecture 10	Embeddings Workshop	HTML, PDF	
Fri Jan 23	Lecture 11	Word Embeddings (Word2Vec, GloVe)	HTML, PDF	Mikolov et al. (2013), Pennington et al. (2014), Assignment 3 Released, Assignment 2 Due

Week 4: Text Embeddings II (January 26–30)

Day	Lecture	Topics	Slides	Materials
Mon Jan 26	Lecture 12	Contextual Embeddings	HTML, PDF	Peters et al. (2018), Cer et al. (2018)
Wed Jan 28	Lecture 13	Dimensionality Reduction	HTML, PDF	McInnes et al. (2018)
Fri Jan 30	Lecture 14	Cognitive Models of Semantic Representation	HTML, PDF	Anderson et al. (2016), Assignment 4 Released, Assignment 3 Due (Feb 2, 11:59 PM EST)

Week 5: Transformers & Attention (February 2–6)

Day	Lecture	Topics	Slides	Materials
Mon Feb 2	Lecture 15	Attention Mechanisms	HTML, PDF	Bahdanau et al. (2015), Vaswani et al. (2017) Vaswani et al. (2017), HuggingFace NLP Course Chapter 3 Assignment 4 Due (Feb 9, 11:59 PM EST)
Wed Feb 4	Lecture 16	Transformer Architecture	HTML, PDF	
Thu Feb 6	Lecture 17	Training Transformers	HTML, PDF	

Week 6: Encoder Models (February 9–13)

Day	Lecture	Topics	Slides	Materials
Mon Feb 9	Lecture 18	BERT Deep Dive	HTML, PDF	Devlin et al. (2019), HuggingFace NLP Course Chapter 4, Assignment 5 Released Liu et al. (2019), Sanh et al. (2019) Assignment 5 Due (Feb 16, 11:59 PM EST)
Wed Feb 11	Lecture 19	BERT Variants	HTML, PDF	
Thu Feb 13	Lecture 20	Applications of Encoder Models	HTML, PDF	

Week 7: Decoder Models & GPT (February 16–20)

Day	Lecture	Topics	Slides	Materials
Mon Feb 16	Lecture 21	GPT Architecture	HTML, PDF	Radford et al. (2018), Radford et al. (2019) Brown et al. (2020), OpenAI (2023)
Wed Feb 18	Lecture 22	Scaling Up to GPT-3 and Beyond	HTML, PDF	
Fri Feb 20	Lecture 23	Implementing GPT from Scratch	HTML, PDF	

Week 8: No Classes (February 23–27)

Instructor away February 23–27. No classes this week. Use this time to work on your final project.

Week 9: Advanced Topics (March 2–6)

Day	Lecture	Topics	Slides	Materials
Mon Mar 2	Lecture 24	Retrieval Augmented Generation (RAG)	HTML, PDF	Lewis et al. (2020)
Wed Mar 4	Lecture 25	Mixture of Experts & Efficiency	HTML, PDF	Fedus et al. (2022), Jiang et al. (2024)
Fri Mar 6	Lecture 26	Ethics, Bias, and Safety	HTML, PDF	Bender et al. (2021)

Week 10: Final Projects (March 9)

Day	Lecture	Topics	Slides	Materials
Mon Mar 9	Lecture 27	Final Project Presentations & Wrap-up	–	Last day of classes, Final Project Due (Mar 9, 11:59 PM EST)

Final Exam Period (March 13–17)

No additional submissions required. Final project materials were due March 9.